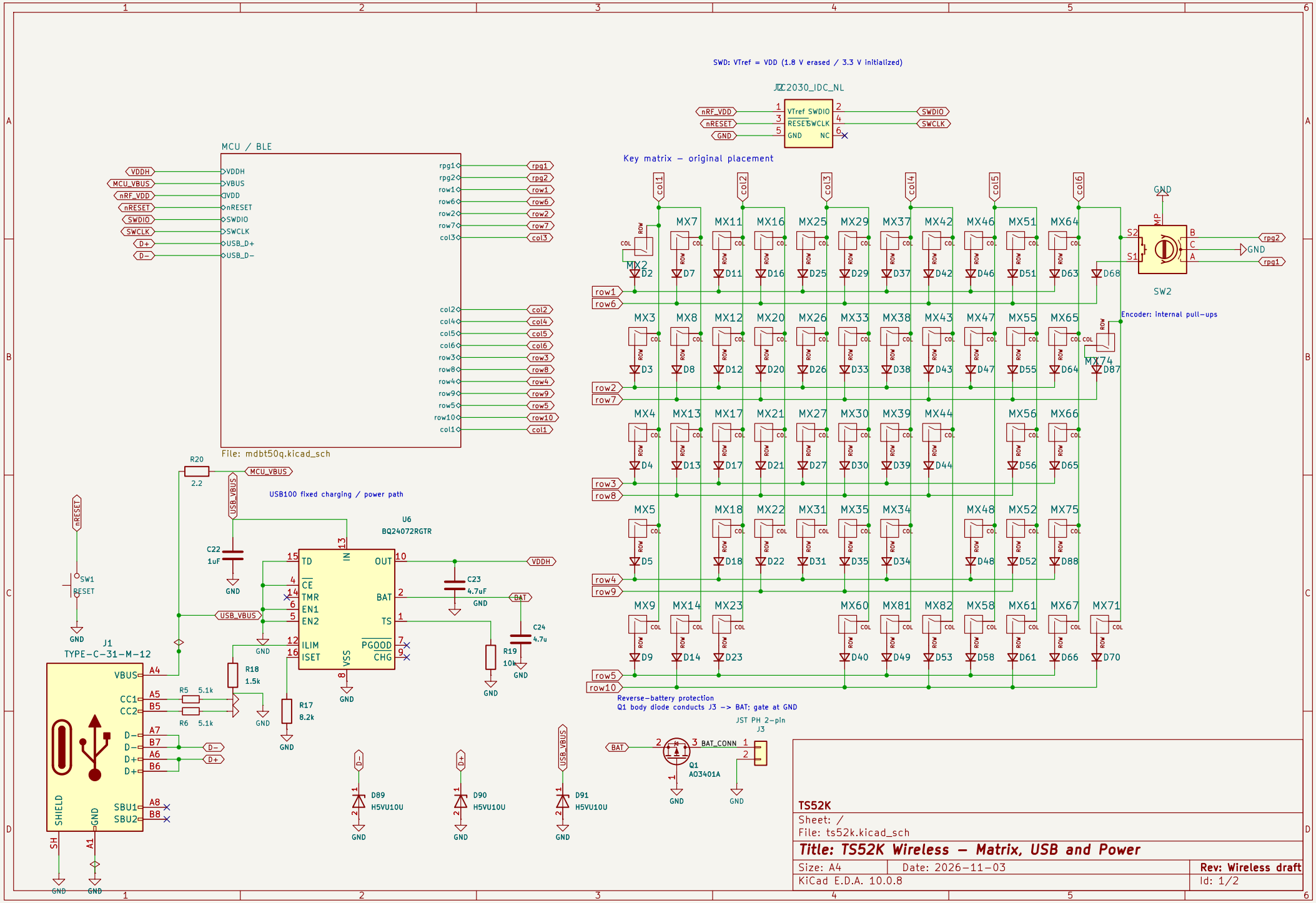
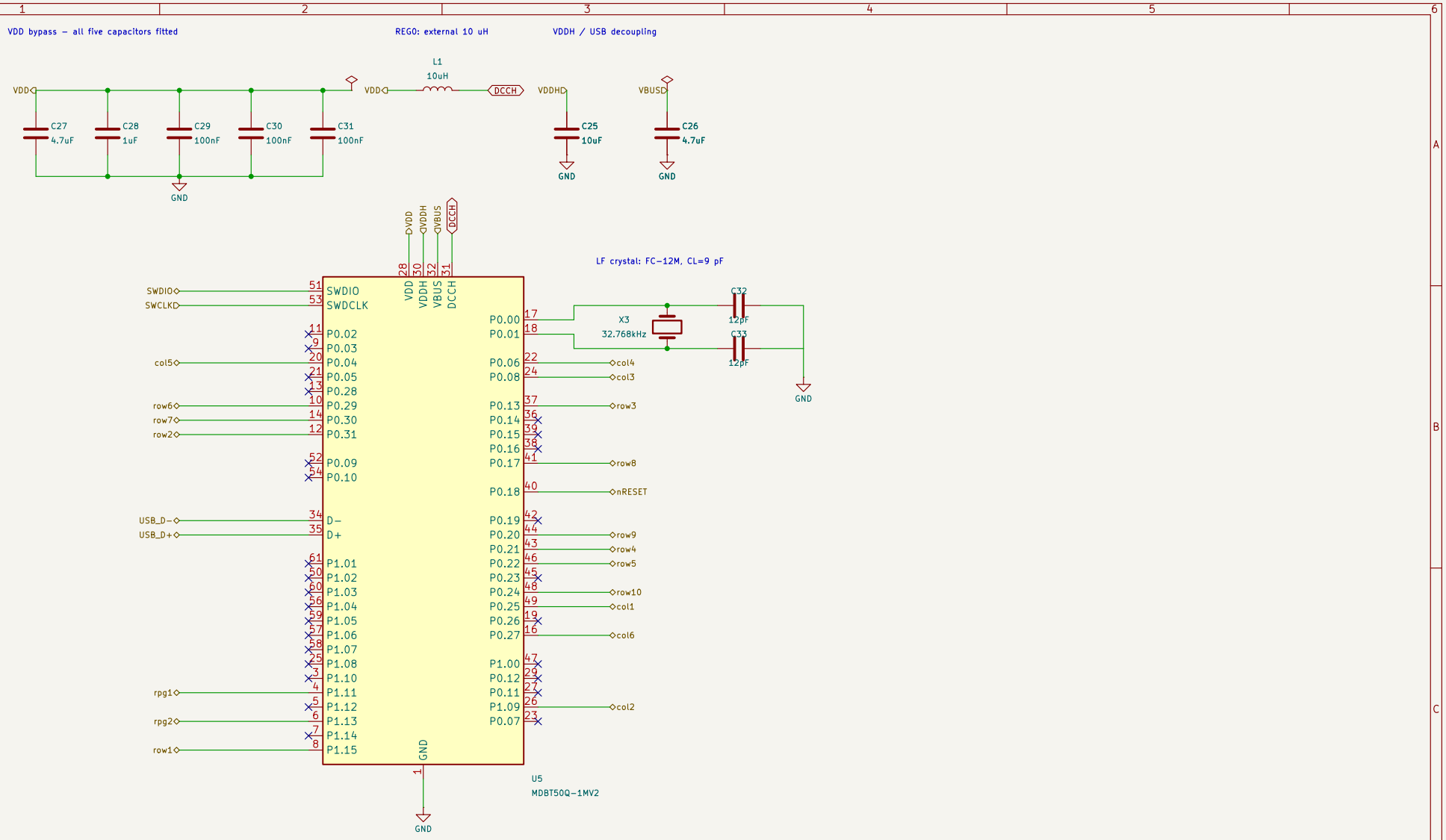

SWD: VTref = VDD (1.8 V erased / 3.3 V initialized)
JTC2030_IDC_NL
nRF_VDD
nRESET
GND
VTref SWDIO
RESET SWCLK
GND NC
SWDIO
SWCLK
MCU / BLE
VDDH
MCU_VBUS
nRF_VDD
nRESET
SWDIO
SWCLK
D+
D−
VDDH
VBUS
VDD
nRESET
SWDIO
SWCLK
USB_D+
USB_D−
rpg1
rpg2
row1
row6
row2
row7
col3
col2
col4
col5
col6
row3
row8
row4
row9
row5
row10
col1
File: mdbt50q.kicad_sch
Key matrix — original placement
col1
col2
col3
col4
col5
col6
MX7
MX11
MX16
MX25
MX29
MX37
MX42
MX46
MX51
MX64
MX2
D2
D7
D11
D16
D25
D29
D37
D42
D46
D51
D63
D68
row1
row6
MX3
MX8
MX12
MX20
MX26
MX33
MX38
MX43
MX47
MX55
MX65
MX74
D3
D8
D12
D20
D26
D33
D38
D43
D47
D55
D64
D87
row2
row7
MX4
MX13
MX17
MX21
MX27
MX30
MX39
MX44
MX56
MX66
D4
D13
D17
D21
D27
D30
D39
D44
D56
D65
row3
row8
MX5
MX18
MX22
MX31
MX35
MX34
MX48
MX52
MX75
D5
D18
D22
D31
D35
D34
D48
D52
D88
row4
row9
MX9
MX14
MX23
MX60
MX81
MX82
MX58
MX61
MX67
MX71
D9
D14
D23
D40
D49
D53
D58
D61
D66
D70
row5
row10
GND
MP
S2
S1
B
C
A
rpg2
GND
rpg1
SW2
Encoder: internal pull-ups
Reverse-battery protection
Q1 body diode conducts J3 -> BAT; gate at GND
JST PH 2-pin
J3
BAT
BAT_CONN
Q1
AO3401A
USB100 fixed charging / power path
U6
BQ24072RGTR
R20
2.2
MCU_VBUS
USB_VBUS
C22
1uF
TD
CE
TMR
EN1
EN2
ILIM
ISET
IN
OUT
BAT
TS
PGOOD
CHG
VSS
VDDH
C23
4.7uF
C24
4.7u
R19
10k
R18
1.5k
R17
8.2k
nRESET
SW1
RESET
J1
TYPE-C-31-M-12
VBUS
CC1
CC2
D−
D+
SHIELD
GND
SBU1
SBU2
SH
A1
R5
5.1k
R6
5.1k
D89
D90
D91
H5VU10U
TS52K
Sheet: /
File: ts52k.kicad_sch
Title: TS52K Wireless — Matrix, USB and Power
Size: A4
Date: 2026-11-03
Rev: Wireless draft
KiCad E.D.A. 10.0.8
Id: 1/2



TS52K		
Sheet: /MCU / BLE/		
File: mdbt50q.kicad_sch		
Title: TS52K Wireless — MDBT50Q and Clock		
Size: A4	Date: 2026-11-03	**Rev: Wireless draft**
KiCad E.D.A. 10.0.8		Id: 2/2